

Movelt Usage

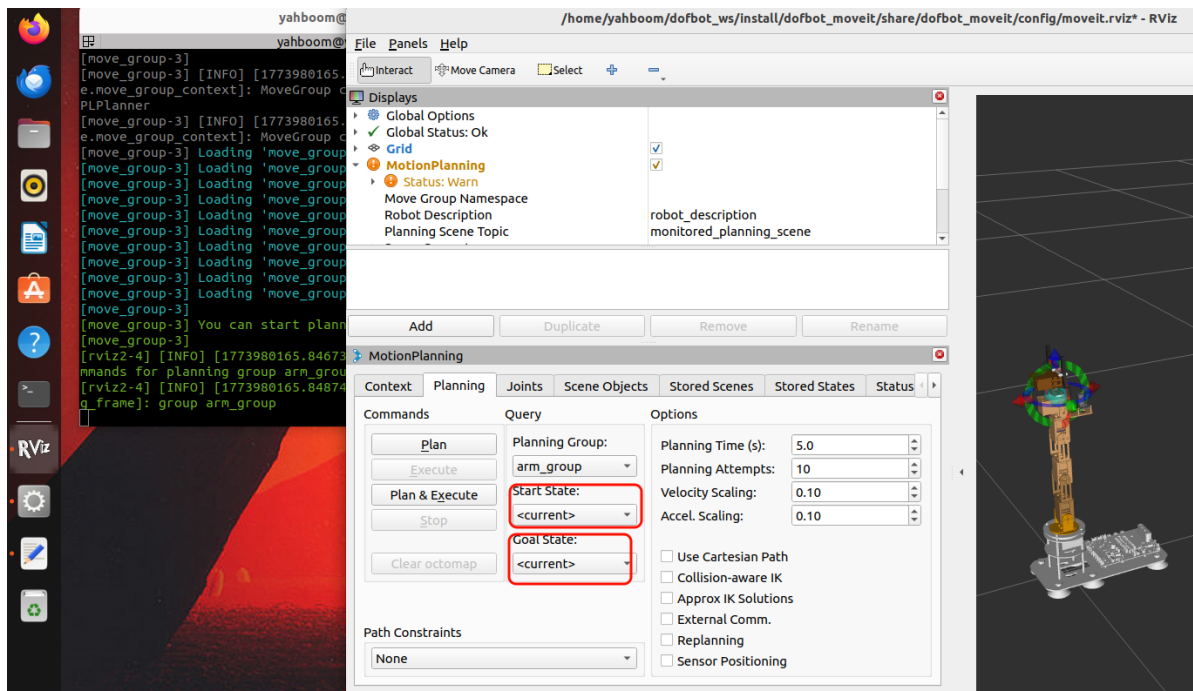
Movelt Usage

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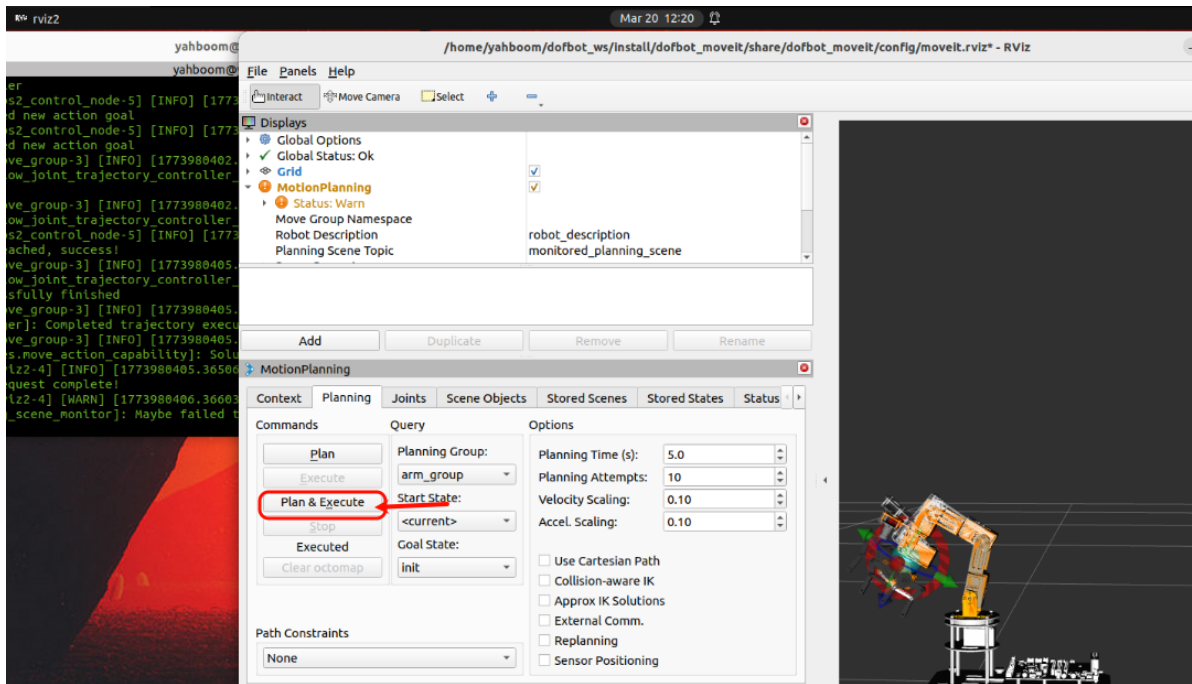
1. Launching Movelt

```
ros2 launch dofbot_moveit demo.launch.py
```

Click the "Planning" tab -> [Start State]: Select "current".



Switch the "Goal State" below to the "Init" state, then click [Plan & Execute]. You should then see the robotic arm change its pose within the simulation environment.

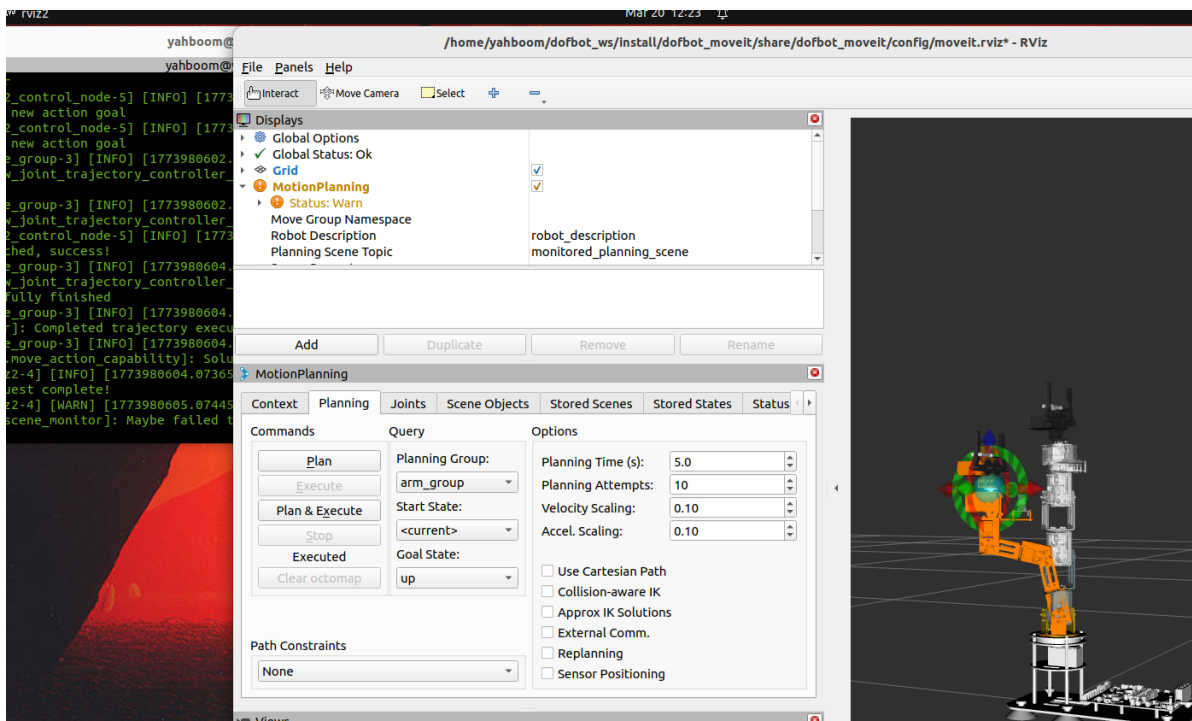


3. Customizing Poses

3.1. Setting the Pose

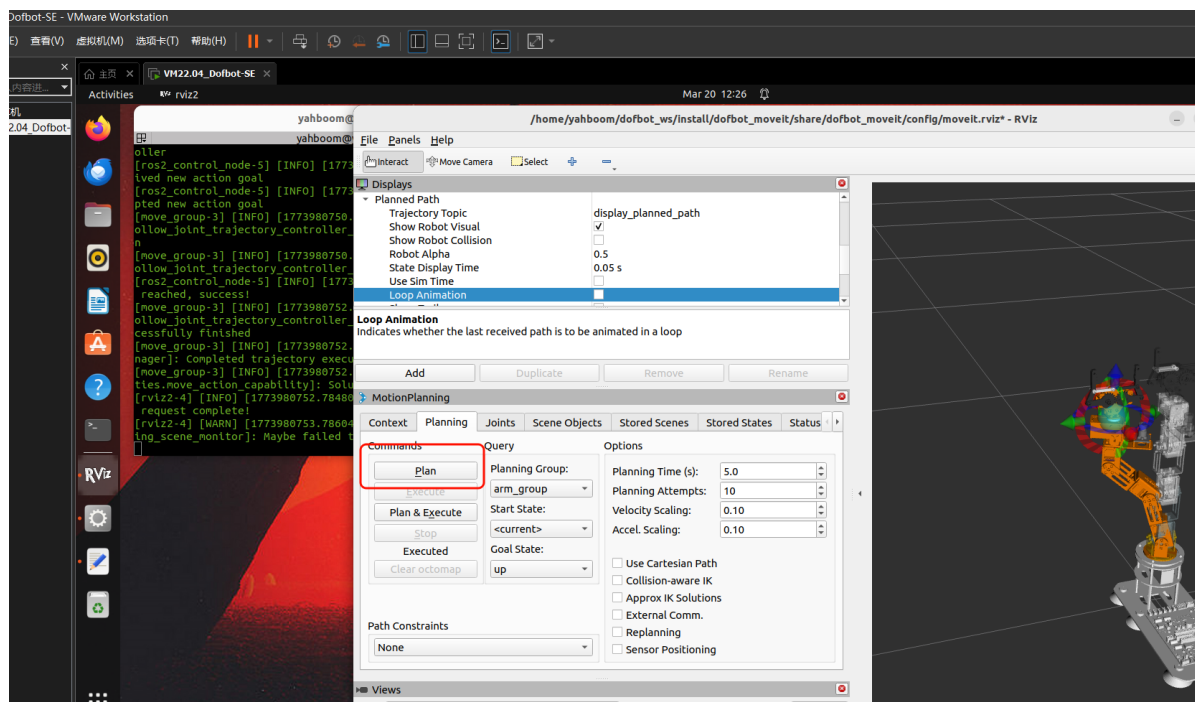
Control the robotic arm's pose by dragging the interactive marker (trackball) located on the arm within RViz. The trackball is the spherical object on the arm; the yellow arm represents the target state.

Note: The robotic arm will only change its pose if the trackball is dragged to a reachable position.



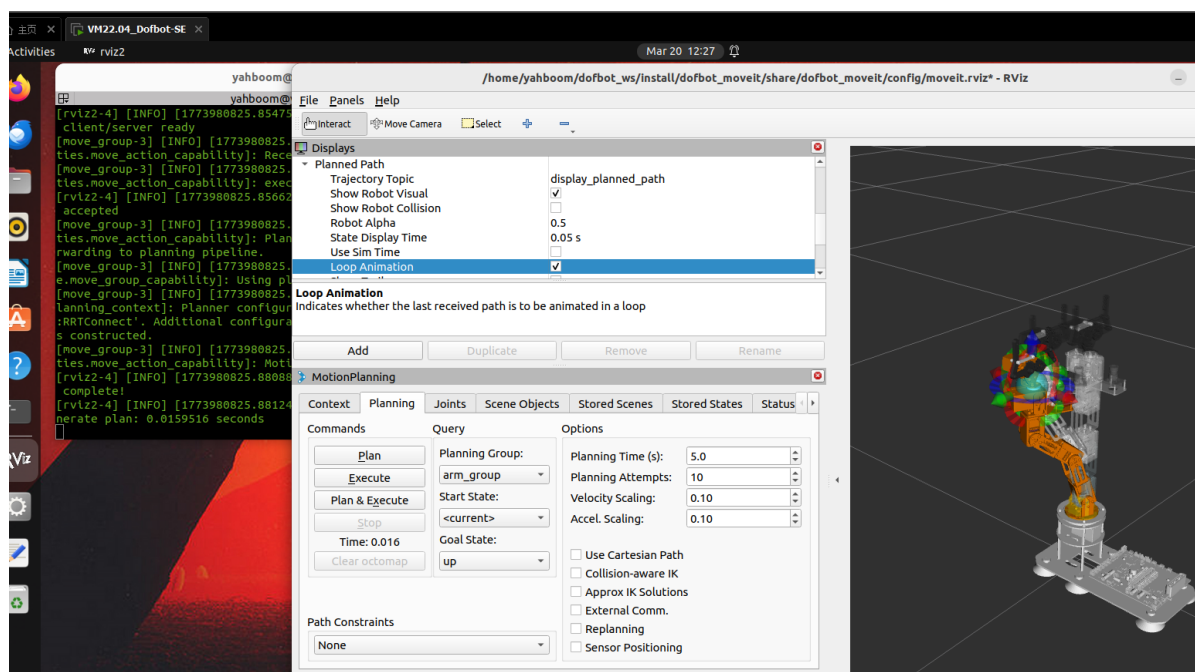
3.2. Planning the Motion

Click **MotionPlanning** → **Planning** → **Plan** to plan the robotic arm's movement to the specified pose.

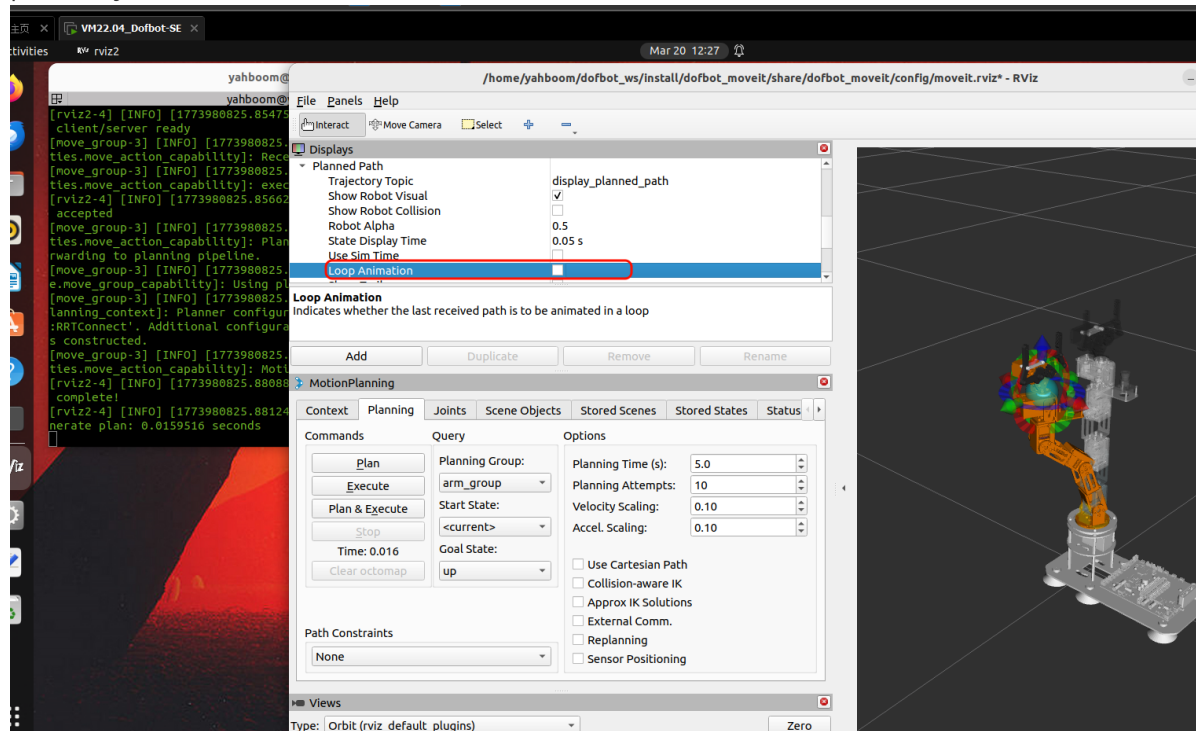


Initiating Motion Planning

The translucent outline of the robotic arm represents the planned motion. If no specific settings have been configured for the planning path, the robotic arm will continuously repeat the planned path.



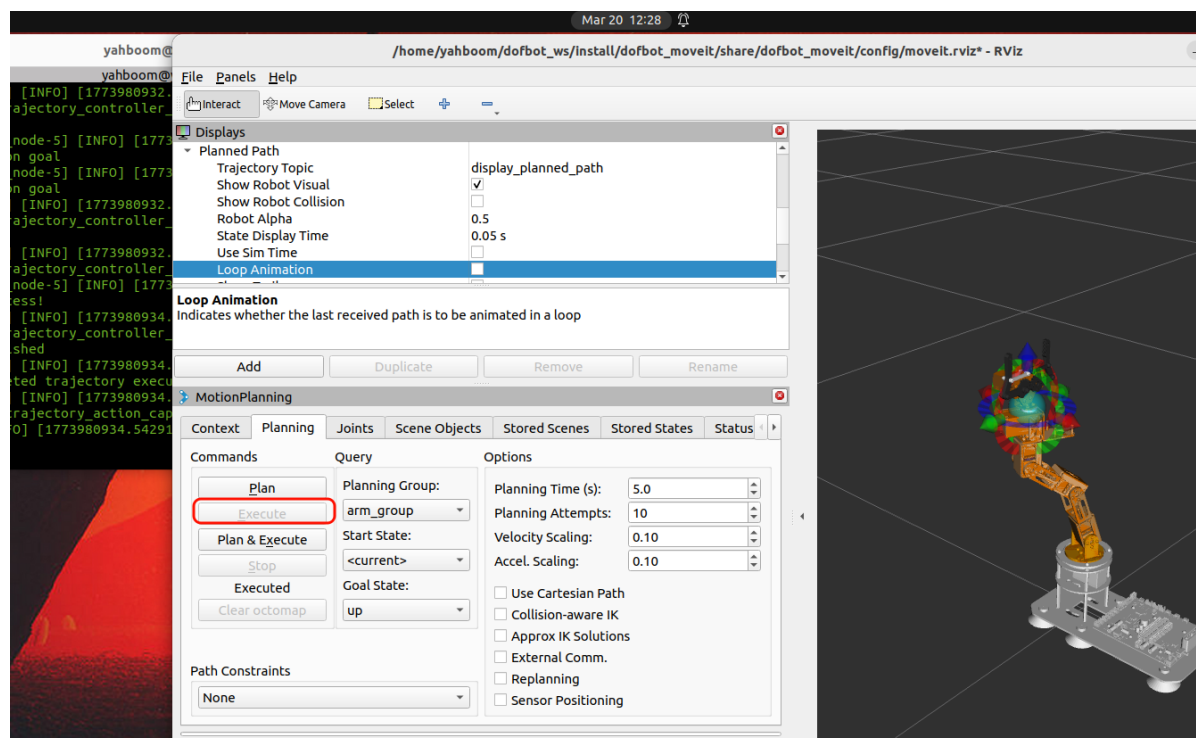
Disable the planning path loop: Once disabled, the robotic arm will plan the path to the specified pose only once.



3.2. Executing the Action

Click **MotionPlanning** → **Planning** → **Execute** to command the robotic arm to move to the specified pose.

Once clicked, the **Execute** button becomes disabled (unclickable) while the planning process is in progress.



4. Preset Poses

Preset poses refer to the various poses for the robotic arm that were configured within the Moveit Setup Assistant.

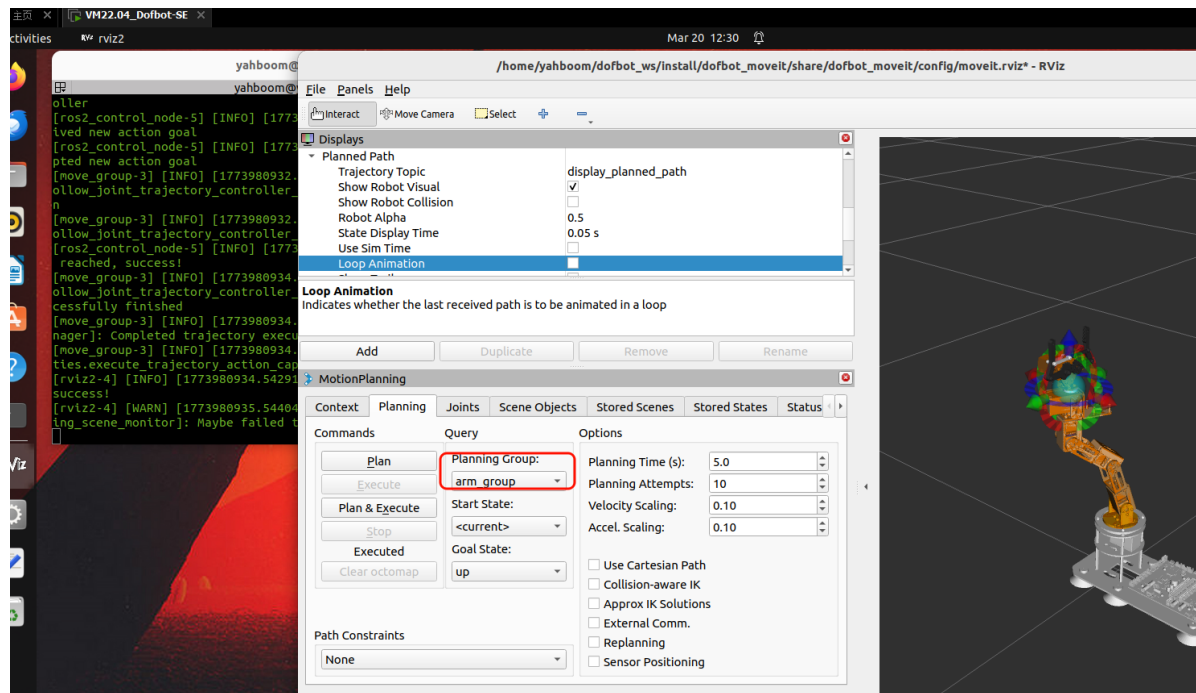
Under **MotionPlanning** → **Planning** → **Planning Group**, you can select the planning group: either the robotic arm or the gripper.

Under **MotionPlanning** → **Planning** → **Start State**, you can set the starting pose for the planning trajectory.

Under **MotionPlanning** → **Planning** → **Goal State**, you can set the target pose for the planning trajectory.

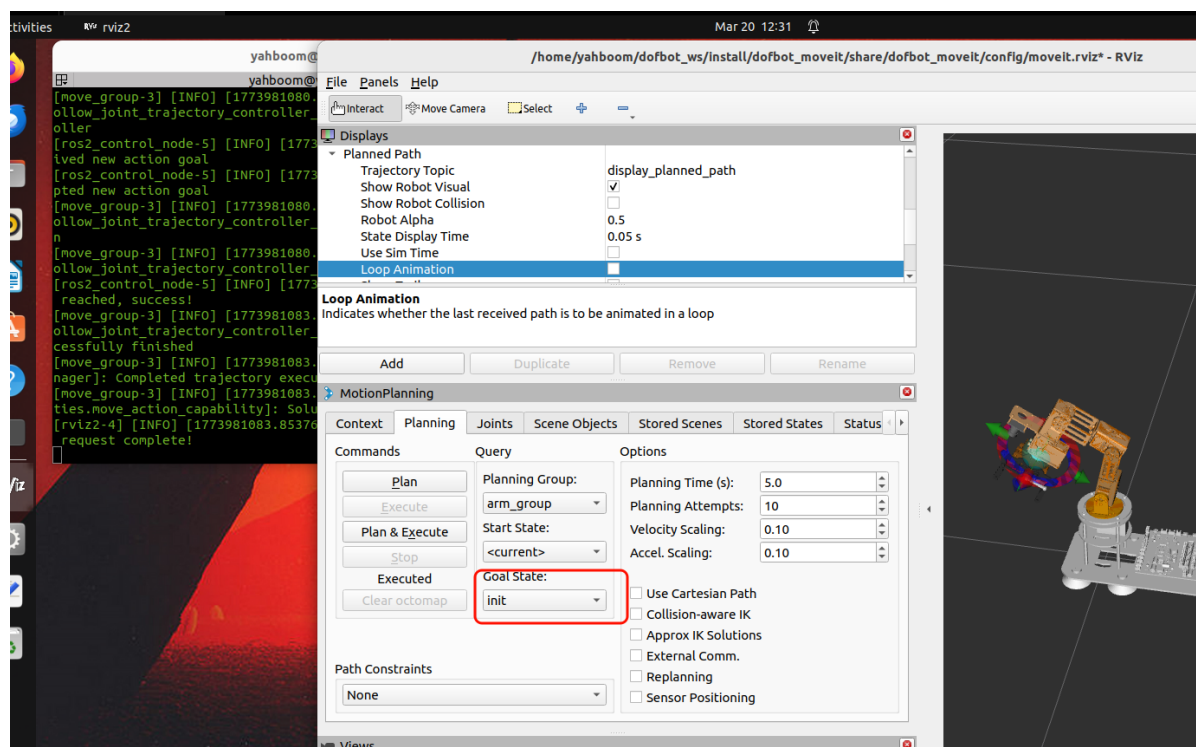
4.1. Robotic Arm Poses

Under **MotionPlanning** → **Planning** → **Planning Group**, set the planning group to: **arm_group**.



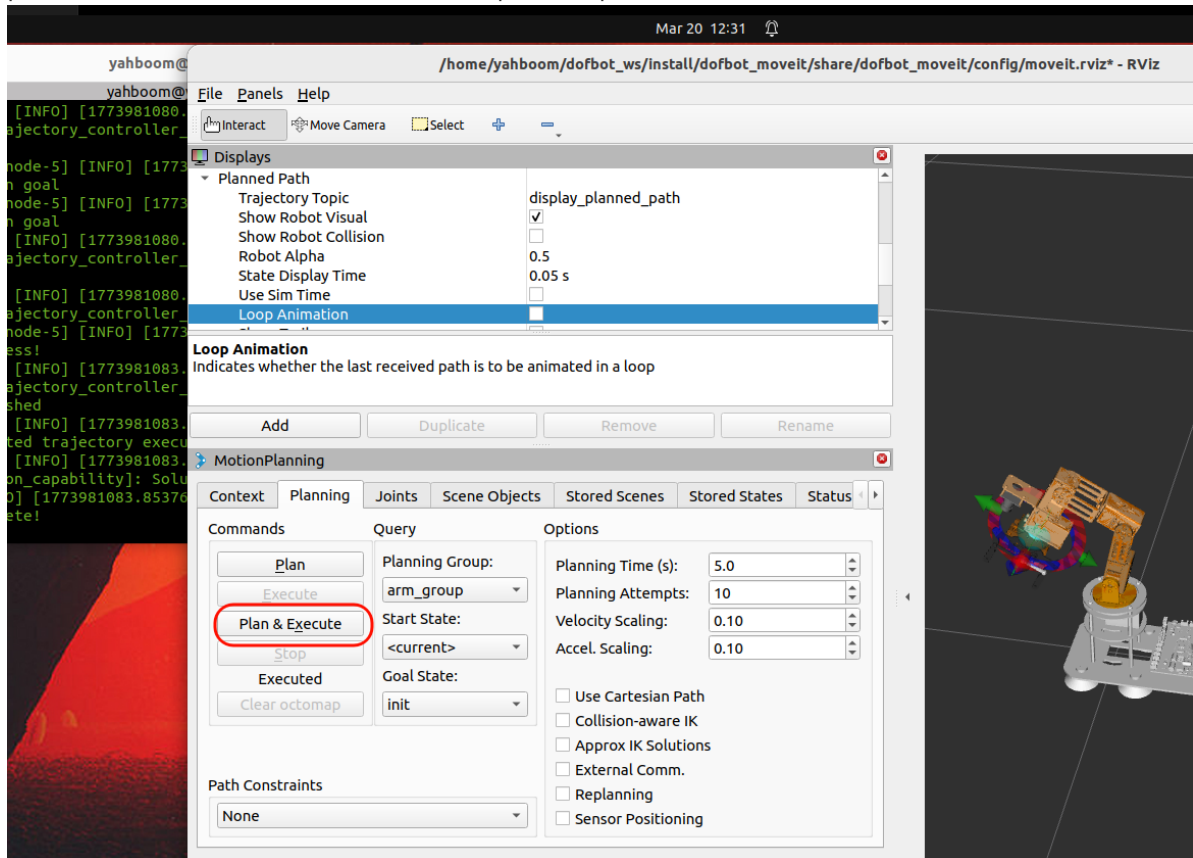
4.1.1. Setting the Pose

Under **MotionPlanning** → **Planning** → **Goal State**, set the target pose for the planning trajectory to: **init**.



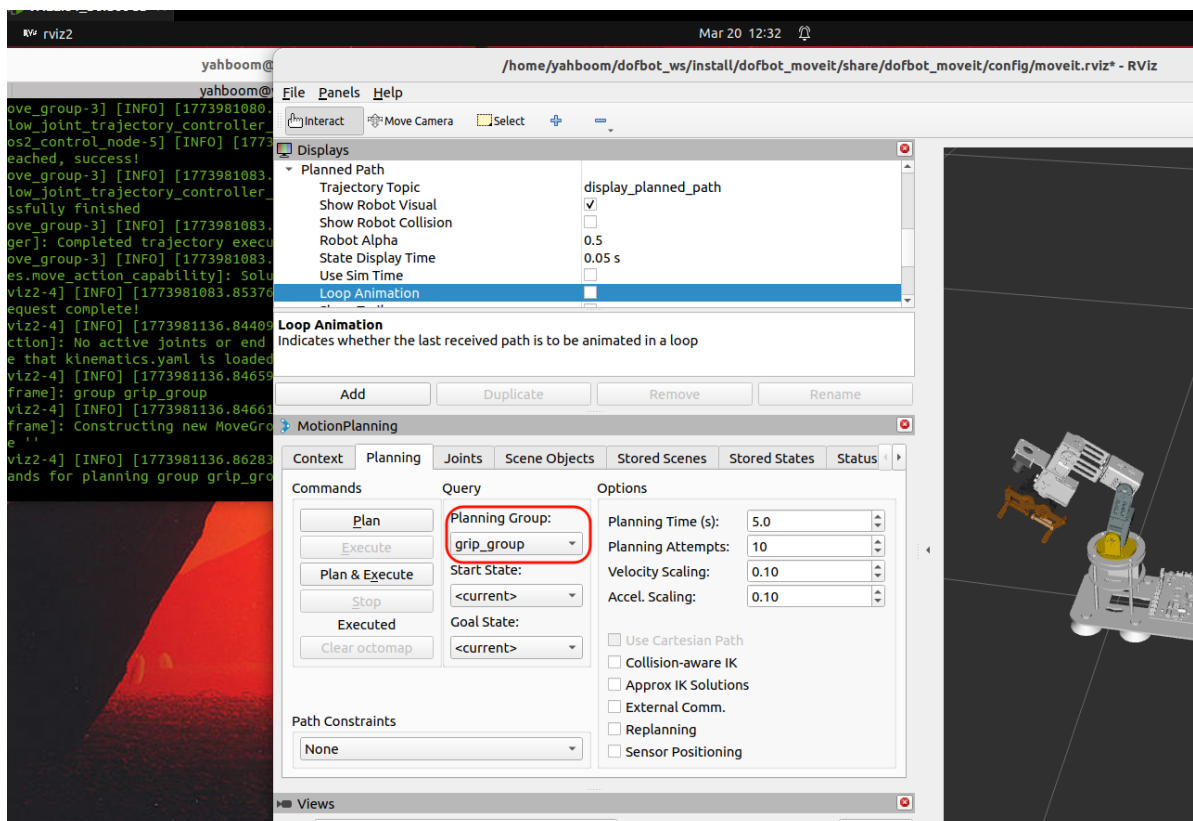
4.1.2. Planning and Executing the Pose

Click **MotionPlanning** → **Planning** → **Plan & Execute** to command the robotic arm to both plan and execute the movement to the specified pose.



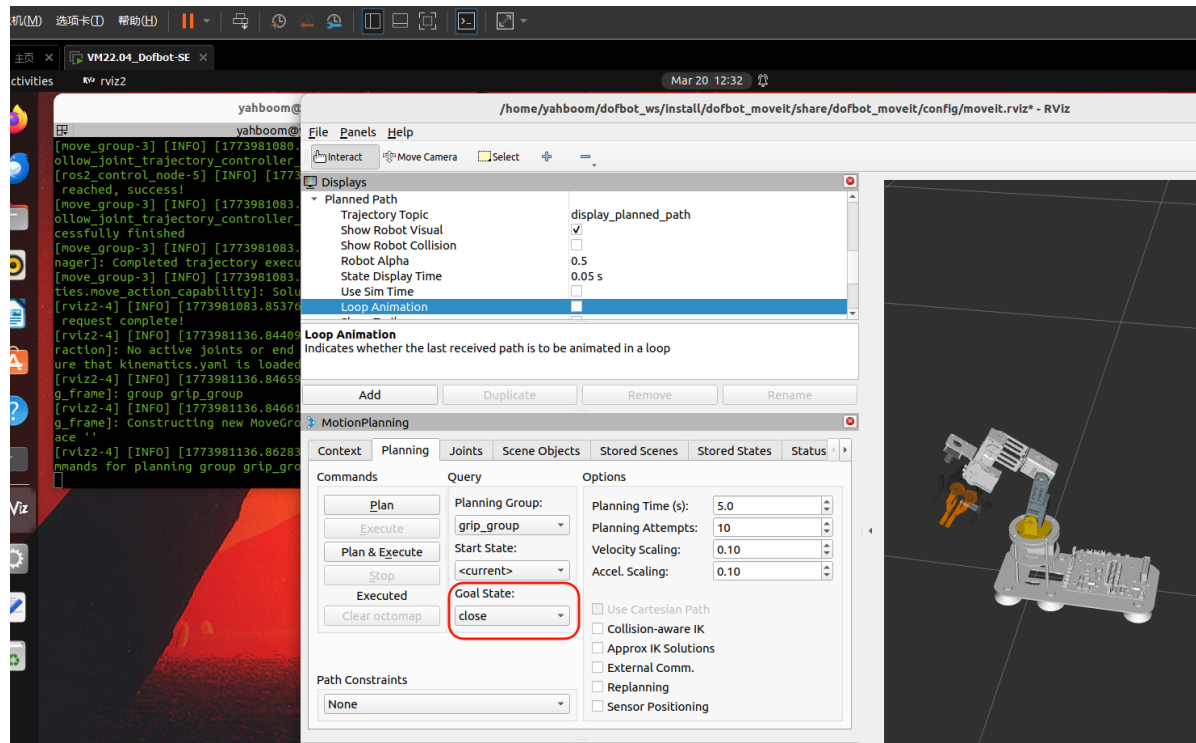
4.2. Gripper Pose

Under **MotionPlanning** → **Planning** → **Planning Group**, set the planning group to: **grip_group**.



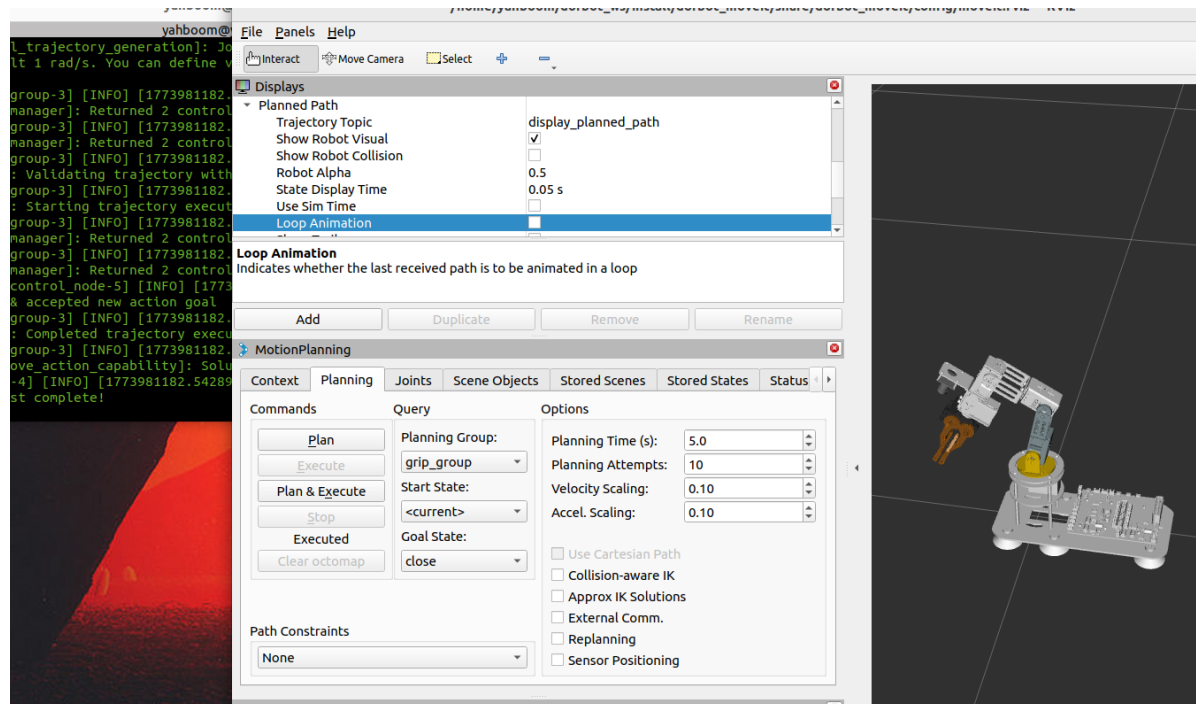
4.2.1. Setting the Pose

Under **MotionPlanning** → **Planning** → **Goal State**, set the target pose for planning to: **close**.



4.2.2. Planning and Executing the Pose

Click **MotionPlanning** → **Planning** → **Plan & Execute** to have the gripper plan and execute the movement to the specified pose.



4.3. Planning Options

MotionPlanning → **Planning** → **Options** allows you to control the Planning Time, Planning Attempts, Velocity Scaling (for planning/execution speed), and Accel. Scaling (for planning/execution acceleration).

Loop Animation

Indicates whether the last received path is to be animated in a loop

AddDuplicateRemoveRename

MotionPlanning

ContextPlanningJointsScene ObjectsStored ScenesStored StatesStatus

Commands

Plan

Execute

Plan & Execute

Stop

Executed

Clear octomap

Query

Planning Group:

grip_group

Start State:

<current>

Goal State:

close

Path Constraints

None

Options

Planning Time (s):5.0

Planning Attempts:10

Velocity Scaling:0.10

Accel. Scaling:0.10

☐ Use Cartesian Path

☐ Collision-aware IK

☐ Approx IK Solutions

☐ External Comm.

☐ Replanning

☐ Sensor Positioning

Views

